# Plane Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Angle relationships

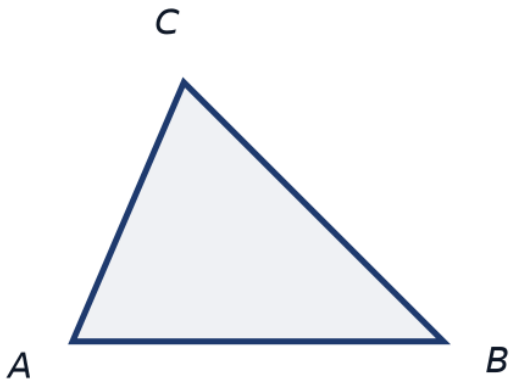
Two angles are supplementary, and one is 3 times the other. Find both angles.

**2** [Foundational] • Target: Angle relationships

Two parallel lines are cut by a transversal, creating an angle of $70^\circ$. Find the measure of its alternate interior angle and its corresponding angle.

**3** [Foundational] • Target: Triangle angle sum and exterior angles

A triangle has angles measuring $x$, $2x$, and $3x$. Find each angle.

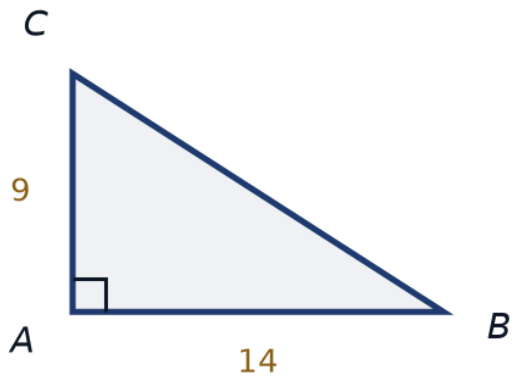



**4** [Foundational] • Target: Triangle angle sum and exterior angles

A triangle's exterior angle at one vertex measures $115^\circ$. If one of the remote interior angles is $50^\circ$, find the other remote interior angle.

**5** [Foundational] • Target: Area and perimeter of polygons

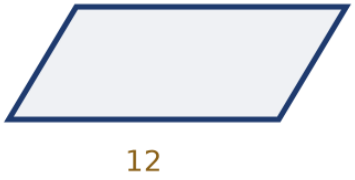
Find the area of a triangle with base 14 and height 9.




© ClipSAT — clipsat.org | All Rights Reserved

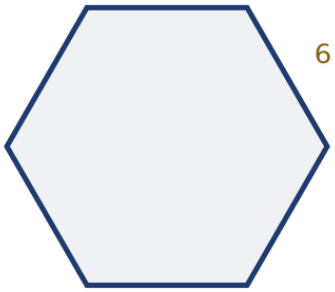
**6** [Foundational] • Target: Area and perimeter of polygons

Find the area of a parallelogram with base 12 and height 5.




**7** [Foundational] • Target: Area and perimeter of polygons

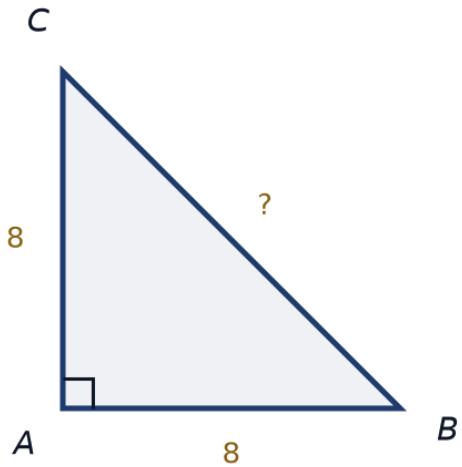
Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.




## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: The Pythagorean theorem and special right triangles

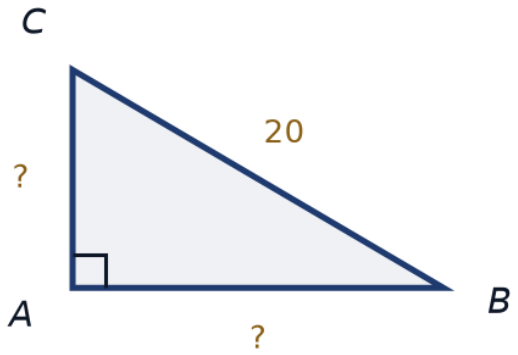
A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.




© ClipSAT — clipsat.org | All Rights Reserved

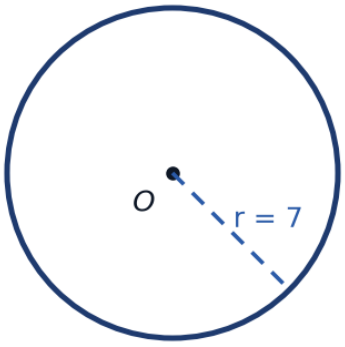
**9** **[Intermediate]** • **Target: The Pythagorean theorem and special right triangles**

A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.




**10** **[Intermediate]** • **Target: Circles: area, circumference, arcs, and sectors**

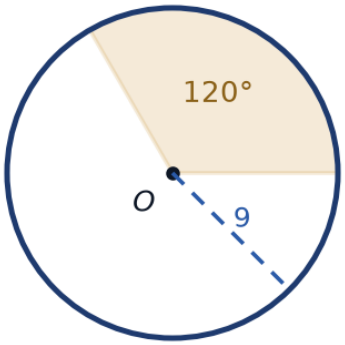
Find the area and circumference of a circle with diameter 14, in terms of $\pi$.




© ClipSAT — clipsat.org | All Rights Reserved

**11** [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

Find the arc length of a $120^\circ$ sector in a circle with radius 9, in terms of $\pi$.

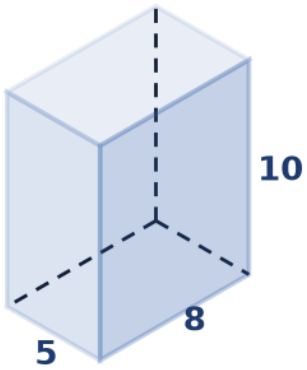



**12** [Intermediate] • Target: Similar and congruent figures

Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

**13** [Intermediate] • Target: Volume and surface area of 3D solids

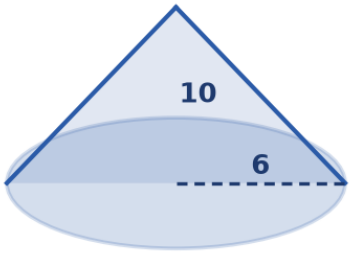
Find the volume of a rectangular prism with dimensions 5, 8, and 10.




© ClipSAT — clipsat.org | All Rights Reserved

**14** [Intermediate] • Target: Volume and surface area of 3D solids

Find the volume of a cone with radius 6 and height 10, in terms of $\pi$.



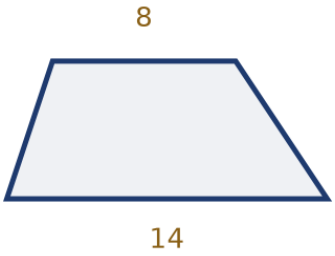

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Multi-step geometry problems • Cross-Lesson: 6.01 Trigonometry

This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of $\pi$.

**16** [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 7.01 Statistics & Probability

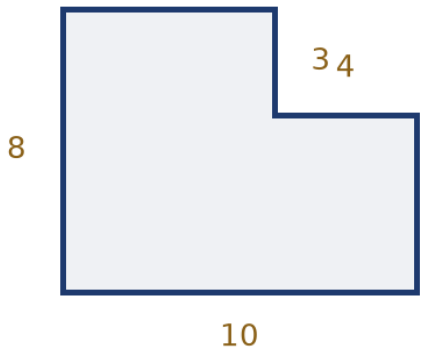
Find the area of a trapezoid with parallel sides 8 and 14, and height 6.




© ClipSAT — clipsat.org | All Rights Reserved

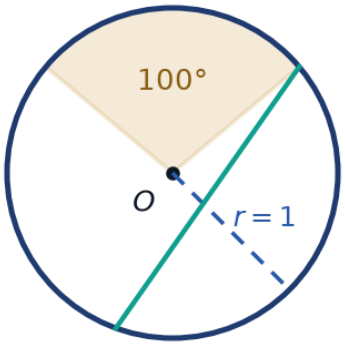
**17** [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.




**18** [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 1.01 Numbers, Percents & Ratios

An inscribed angle subtends an arc of $100^\circ$. Find the measure of the inscribed angle.




**19** [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A tangent line touches a circle at point $P$. Explain the relationship between the tangent line and the radius drawn to point $P$.

**20** [Challenging] • Target: Congruence and triangle proofs • Cross-Lesson: 3.01 Functions & Graphs

State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

© ClipSAT — clipsat.org | All Rights Reserved

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Angle relationships

Two angles are complementary, and they differ by $52^\circ$. Find the larger angle.

**a.** $19^\circ$

**b.** $66^\circ$

**c.** $76^\circ$

**d.** $71^\circ$

**22** [Foundational] • Target: Triangle angle sum and exterior angles

Two angles are supplementary, and they differ by $4^\circ$. Find the larger angle.

**a.** $87^\circ$

**b.** $88^\circ$

**c.** $97^\circ$

**d.** $92^\circ$

**23** [Foundational] • Target: Area and perimeter of polygons

A right triangle has legs 12 and 16. Find the length of the hypotenuse.

**a.** 28

**b.** 21

**c.** 19

**d.** 20

**24** [Foundational] • Target: The Pythagorean theorem and special right triangles

A circle has radius 11. Find the arc length of a $270^\circ$ sector, in terms of $\pi$.

**a.** $\frac{33}{2}\pi$

**b.** $\frac{35}{2}\pi$

**c.** $\frac{31}{2}\pi$

**d.** $33\pi$

**25** [Foundational] • Target: Circles: area, circumference, arcs, and sectors

A circle has radius 2. Find the sector area of a $180^\circ$ sector, in terms of $\pi$.

**a.** $2\pi$

**b.** $3\pi$

**c.** $4\pi$

**d.** $1\pi$

© ClipSAT — clipsat.org | All Rights Reserved

**26** [Foundational] • Target: Similar and congruent figures

Find the volume of a cylinder with radius 5 and height 14, in terms of $\pi$.

**a.** $350\pi$

**b.** $360\pi$

**c.** $140\pi$

**d.** $340\pi$

**27** [Foundational] • Target: Volume and surface area of 3D solids

In a triangle, two remote interior angles measure $50^\circ$ and $50^\circ$. What is the exterior angle at the third vertex?

**a.** $100^\circ$

**b.** $45^\circ$

**c.** $53^\circ$

**d.** $178^\circ$

**28** [Foundational] • Target: Multi-step geometry problems

Two similar triangles have a scale factor of 3. If the smaller triangle has area 9, what is the area of the larger triangle?

**a.** 136

**b.** 86

**c.** 147

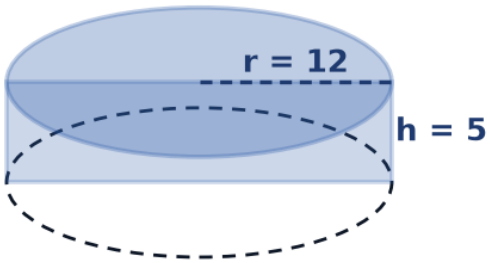
**d.** 81

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

**29** [Intermediate] • Target: Trapezoids and composite polygon areas

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?




**a.** $100\pi$

**b.** $\frac{400\pi}{3}$

**c.** $\frac{500\pi}{3}$

**d.** $200\pi$

**30** [Intermediate] • Target: Chords, tangents, and inscribed angles in circles

What is each exterior angle of a regular 12-sided polygon?

**a.** $54^\circ$

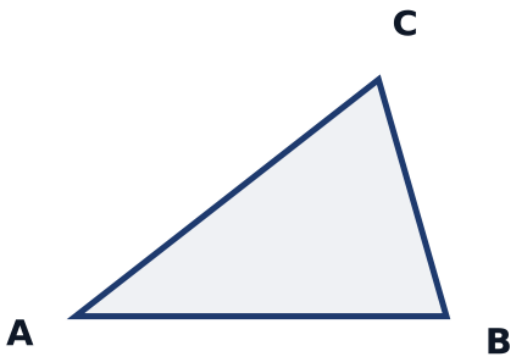
**b.** $42^\circ$

**c.** $30^\circ$

**d.** $4^\circ$

**31** [Intermediate] • Target: Congruence and triangle proofs

In a triangle, two of the angles measure $38^\circ$ and $74^\circ$. What is the measure of the third angle?




**a.** 64°

**b.** 68°

**c.** 66°

**d.** 59°

© ClipSAT — clipsat.org | All Rights Reserved

**32** **[Intermediate]** • **Target: A multi-part composite geometry problem**

Two angles of a triangle measure $81^\circ$ and $54^\circ$. What is the third angle?

**a.** $39^\circ$

**b.** $45^\circ$

**c.** $171^\circ$

**d.** $1^\circ$

**33** **[Intermediate]** • **Target: Angle relationships**

What is the sum of the interior angles of a 5-sided polygon?

**a.** $879^\circ$

**b.** $540^\circ$

**c.** $430^\circ$

**d.** $75^\circ$

**34** **[Intermediate]** • **Target: Triangle angle sum and exterior angles**

Two similar triangles have a scale factor of 5. If a side of the smaller triangle is 5, what is the corresponding side of the larger triangle?

**a.** 37

**b.** 36

**c.** 25

**d.** 19

**35** **[Intermediate]** • **Target: Area and perimeter of polygons**

In a right triangle with acute angles $A$ and $B$, $\sin(A) = \frac{5}{13}$. What is the value of $\cos(B) + \tan(A)$?

**a.** $\frac{5}{13} + \frac{5}{12} = \frac{125}{156}$

**b.** $\frac{12}{13} + \frac{5}{12}$

**c.** $\frac{5}{13} + \frac{12}{5}$

**d.** $1$

**36** **[Intermediate]** • **Target: The Pythagorean theorem and special right triangles**

A right triangle has area 24 and hypotenuse 10. What is the perimeter of the triangle?

**a.** 20

**b.** 22

**c.** 24

**d.** 26

© ClipSAT — clipsat.org | All Rights Reserved

**37** [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

A right triangle has hypotenuse 10 and one acute angle measuring $30^\circ$. What is the area of the triangle?

**a.** $12.5\sqrt{3}$

**b.** $25\sqrt{3}$

**c.** 25

**d.** 50

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Similar and congruent figures • Cross-Lesson: 1.01 Numbers, Percents & Ratios

In a right triangle with acute angle $\theta$, the side opposite $\theta$ is 9 and the hypotenuse is 15. What is $\sin(\theta)$?

**a.** $\frac{3}{5}$

**b.** $\frac{4}{5}$

**c.** $\frac{3}{4}$

**d.** $\frac{5}{3}$

**39** [Challenging] • Target: Volume and surface area of 3D solids • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In a right triangle, acute angle $A$ measures $30^\circ$. What is $\sin(30^\circ)$?

**a.** $\frac{1}{2}$

**b.** $\frac{\sqrt{2}}{2}$

**c.** $\frac{\sqrt{3}}{2}$

**d.** 1

**40** [Challenging] • Target: Multi-step geometry problems • Cross-Lesson: 3.01 Functions & Graphs

In a right triangle, the side opposite angle $\theta$ has length $8$, the side adjacent to $\theta$ has length $15$, and the hypotenuse has length $17$. What is $cos(\theta)$?

**a.** 77/68

**b.** 283/340

**c.** 92/85

**d.** 15/17

**41** [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 4.01 Coordinate Geometry

In a right triangle, the length of the hypotenuse is 12, and the sum of the lengths of the two legs is 16. If $\theta$ is the measure of one of the acute angles of the triangle, what is the value of $\sin(2\theta)$?

**a.** $\frac{7}{18}$

**b.** $\frac{7}{16}$

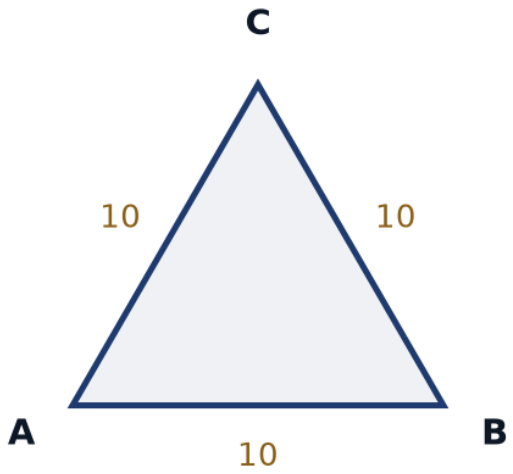
**c.** $\frac{5}{9}$

**d.** $\frac{7}{9}$

© ClipSAT — clipsat.org | All Rights Reserved

**42** [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 6.01 Trigonometry

An equilateral triangle has sides of length 10. What is its perimeter?




**a.** 20

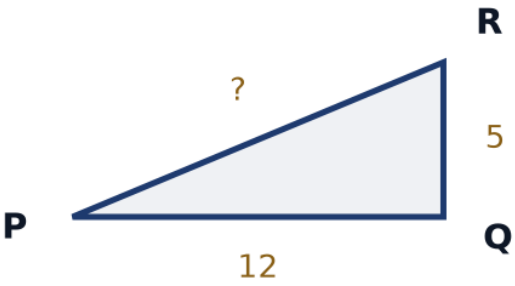
**b.** 25

**c.** 30

**d.** 35

**43** [Challenging] • Target: Congruence and triangle proofs • Cross-Lesson: 7.01 Statistics & Probability

A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?




**a.** 13

**b.** 15

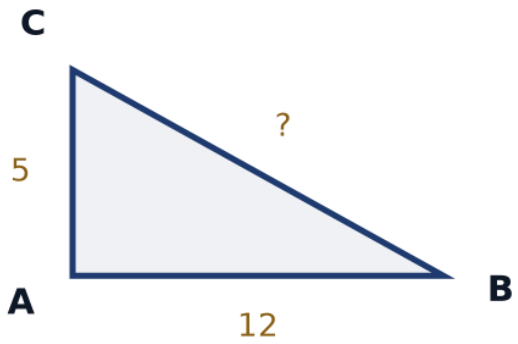
**c.** 17

**d.** $\sqrt{119}$

© ClipSAT — clipsat.org | All Rights Reserved

**44** [Challenging] • Target: A multi-part composite geometry problem • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

In a right triangle, the legs are 5 and 12. What is the hypotenuse?




**a.** 13

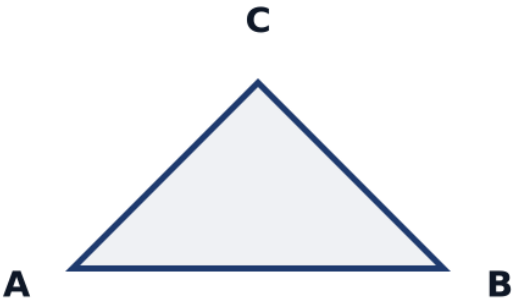
**b.** 15

**c.** 17

**d.** $\sqrt{119}$

**45** [Challenging] • Target: Angle relationships • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If two angles in a triangle are $45^\circ$ and $45^\circ$, what type of triangle is it?




**a.** Acute equilateral

**b.** Right isosceles

**c.** Obtuse isosceles

**d.** Right scalene

© ClipSAT — clipsat.org | All Rights Reserved